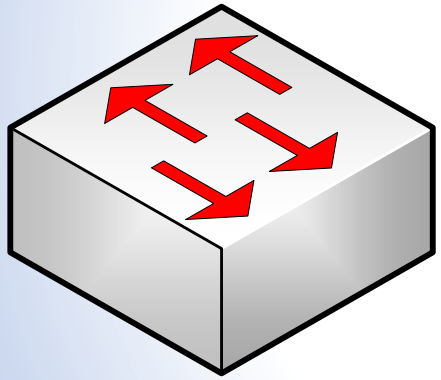
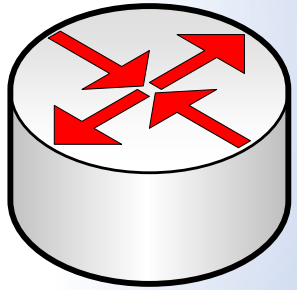
Switch vs. Router



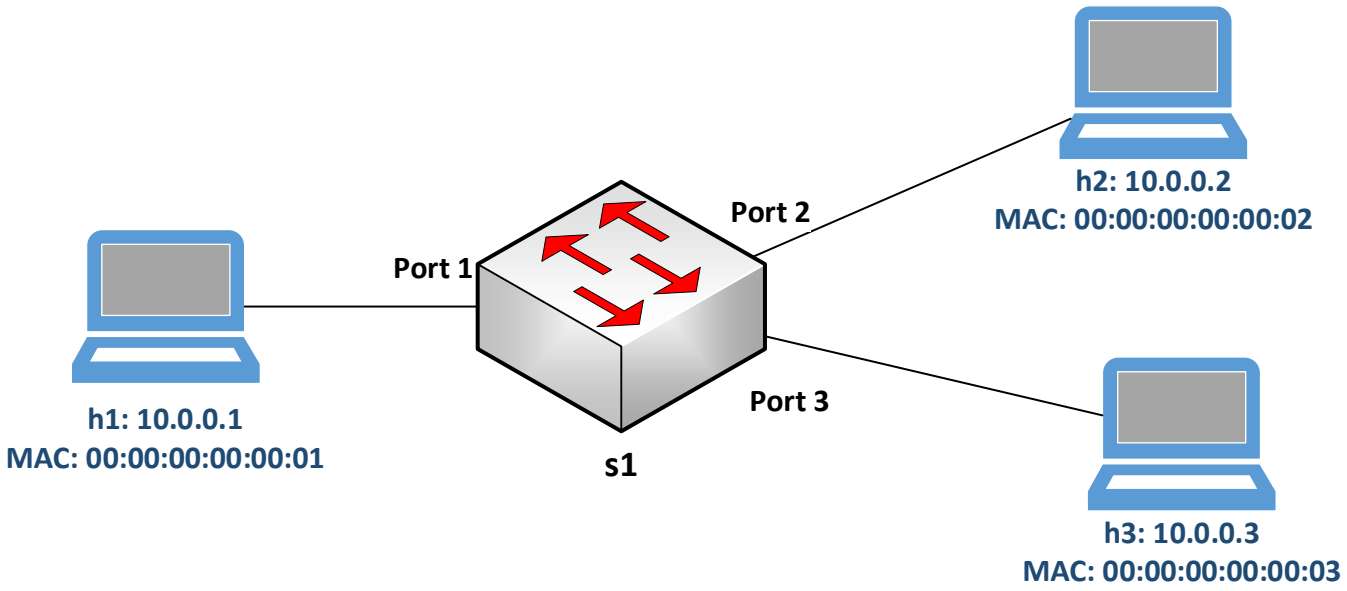




Diferencias

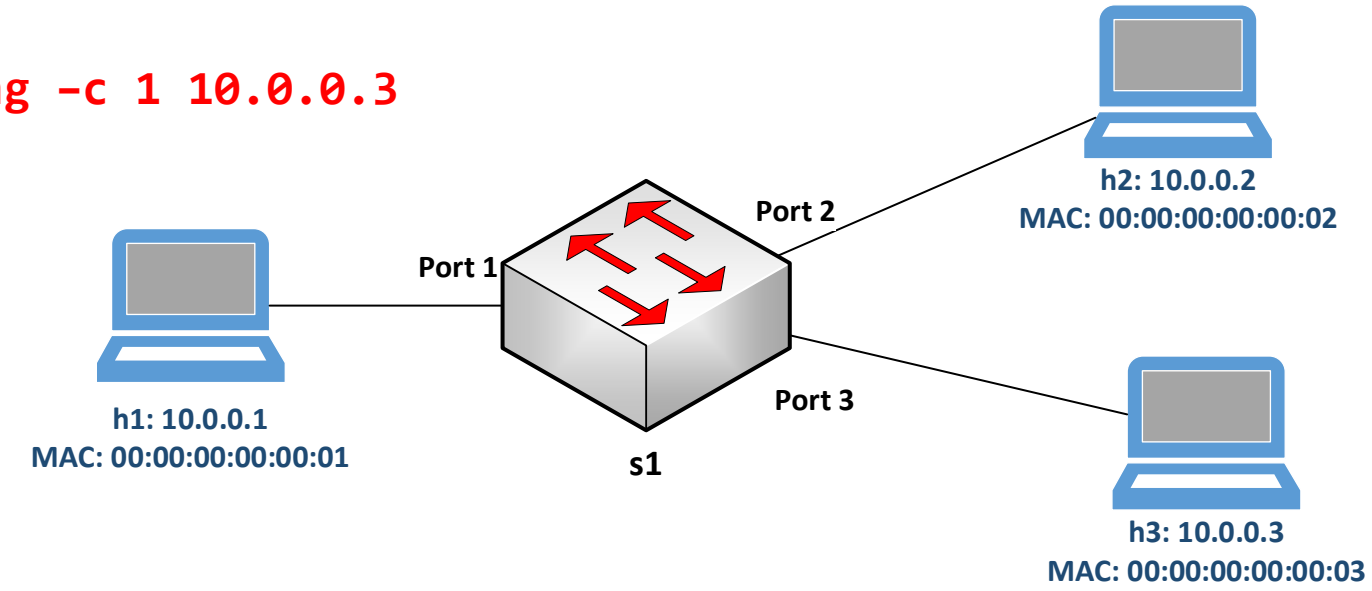
	Switch	Router
Capa	2 (Data Link)	3 (Red)
Qué direcciones mira?	MAC	IP
Qué conecta?	Dispositivos de una misma red local (LAN)	Redes distintas
Cómo reenvía los datos?	Aprende direcciones MAC	Usa una Tabla de Ruteo
Qué divide?	Dominios de colisión	Dominios de Broadcast
Cantidad de puertos (casos típicos)	Muchos. 16, 24, 48 o más (modulares)	Pocos. Menos de 8 en Gral. (excepto modulares de grandes ISPs)

Topología Simple: Switch



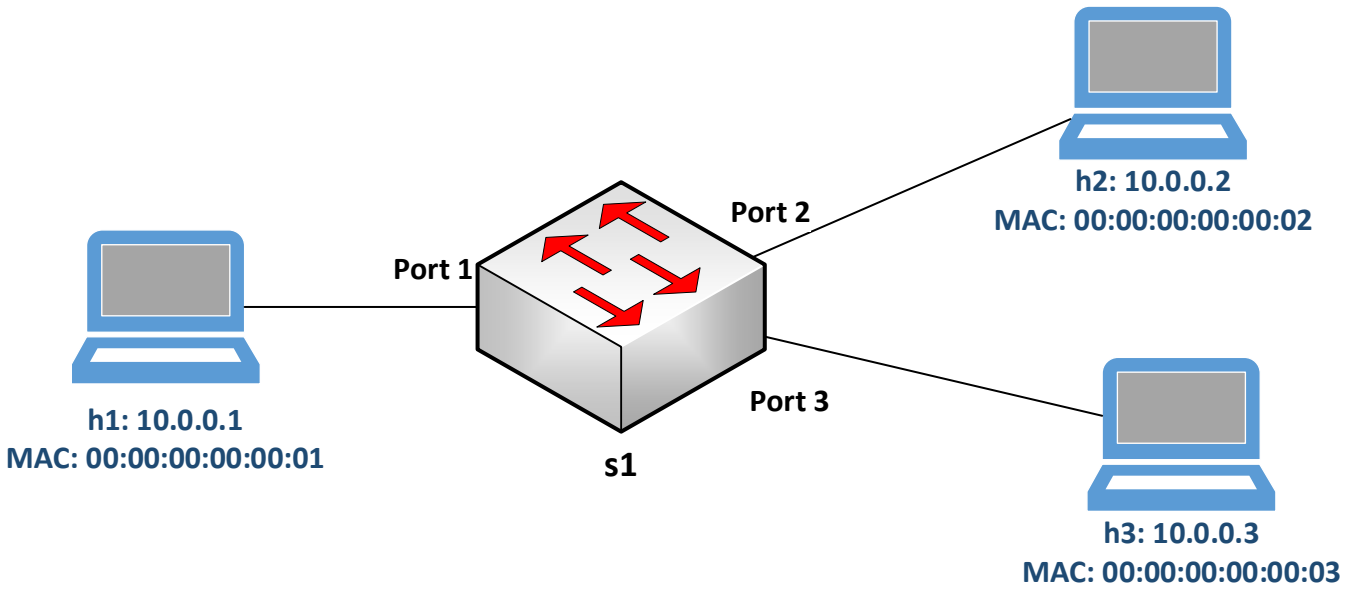
Ping

```
h1:~$ping -c 1 10.0.0.3
```



Tablas (Vacías)

```
h1:~$ping -c 1 10.0.0.3
```



Switch Table	
MAC	Port

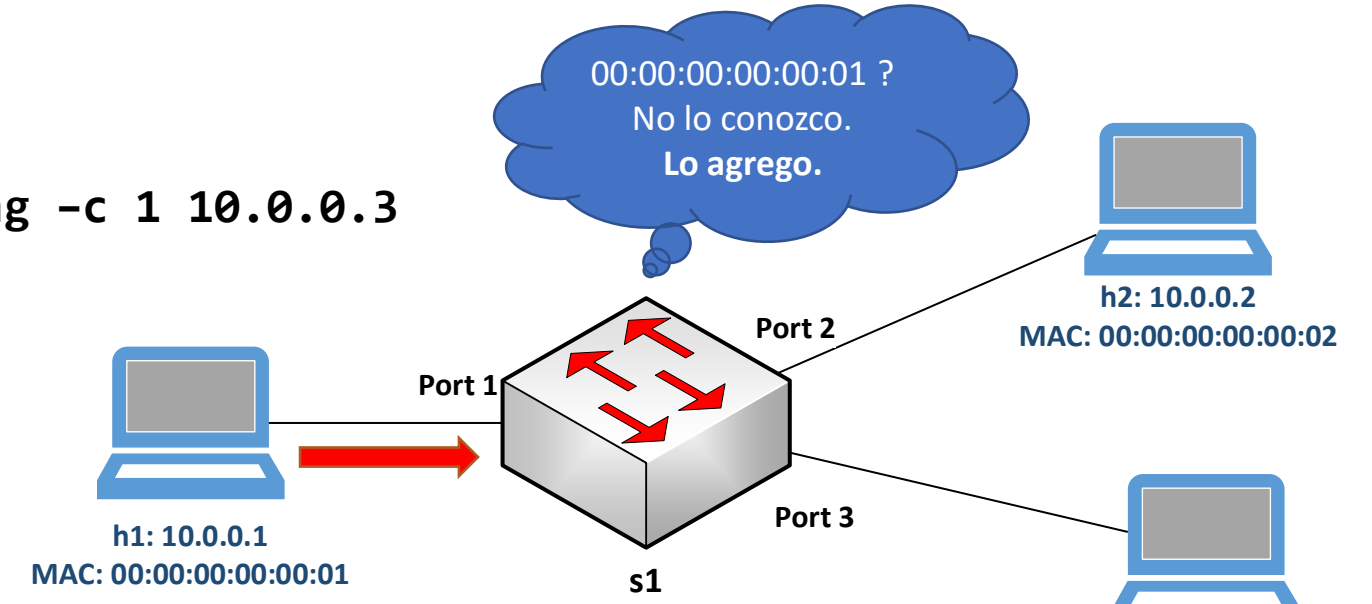
h1 ARP Table	
IP	MAC

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC

ARP Request

```
h1:~$ping -c 1 10.0.0.3
```



h1 → ARP Request → s1:Port1
10.0.0.1 → 10.0.0.3
00:00:00:00:00:01 → FF:FF:FF:FF:FF:FF

Switch Table	
MAC	Port
00:00:00:00:00:01	1

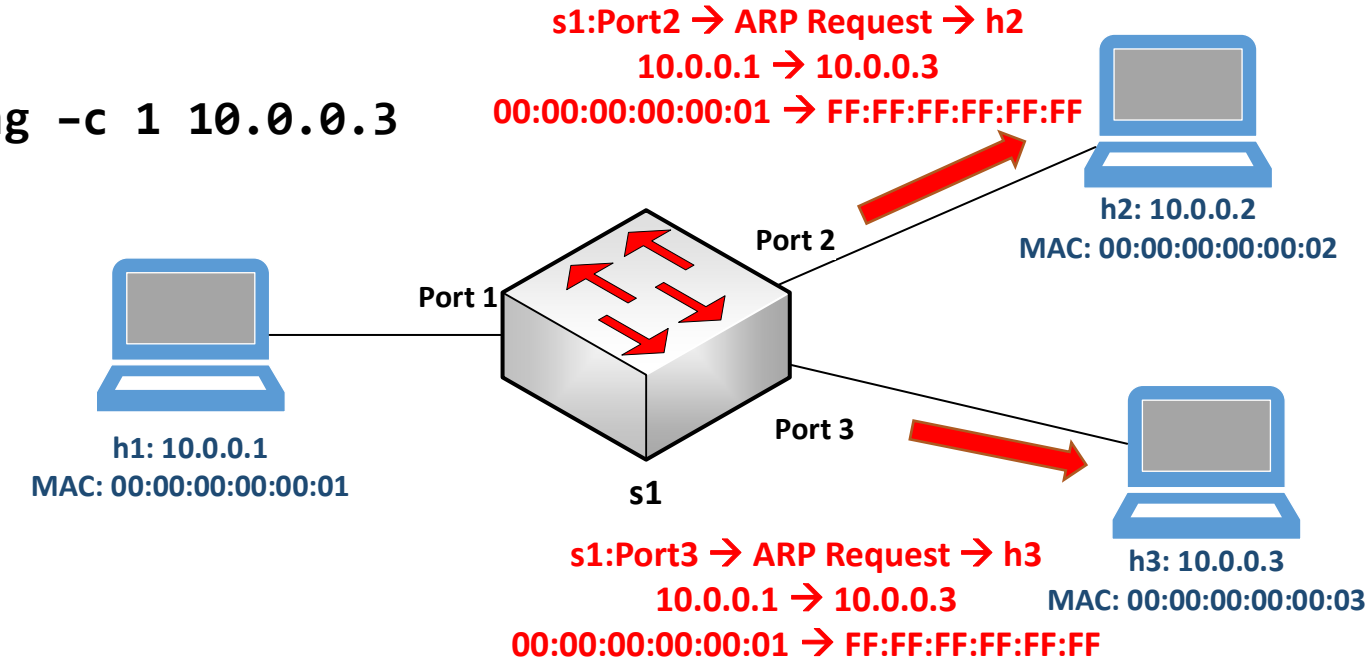
h1 ARP Table	
IP	MAC

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC

Flooding

```
h1:~$ping -c 1 10.0.0.3
```



Switch Table	
MAC	Port
00:00:00:00:00:01	1

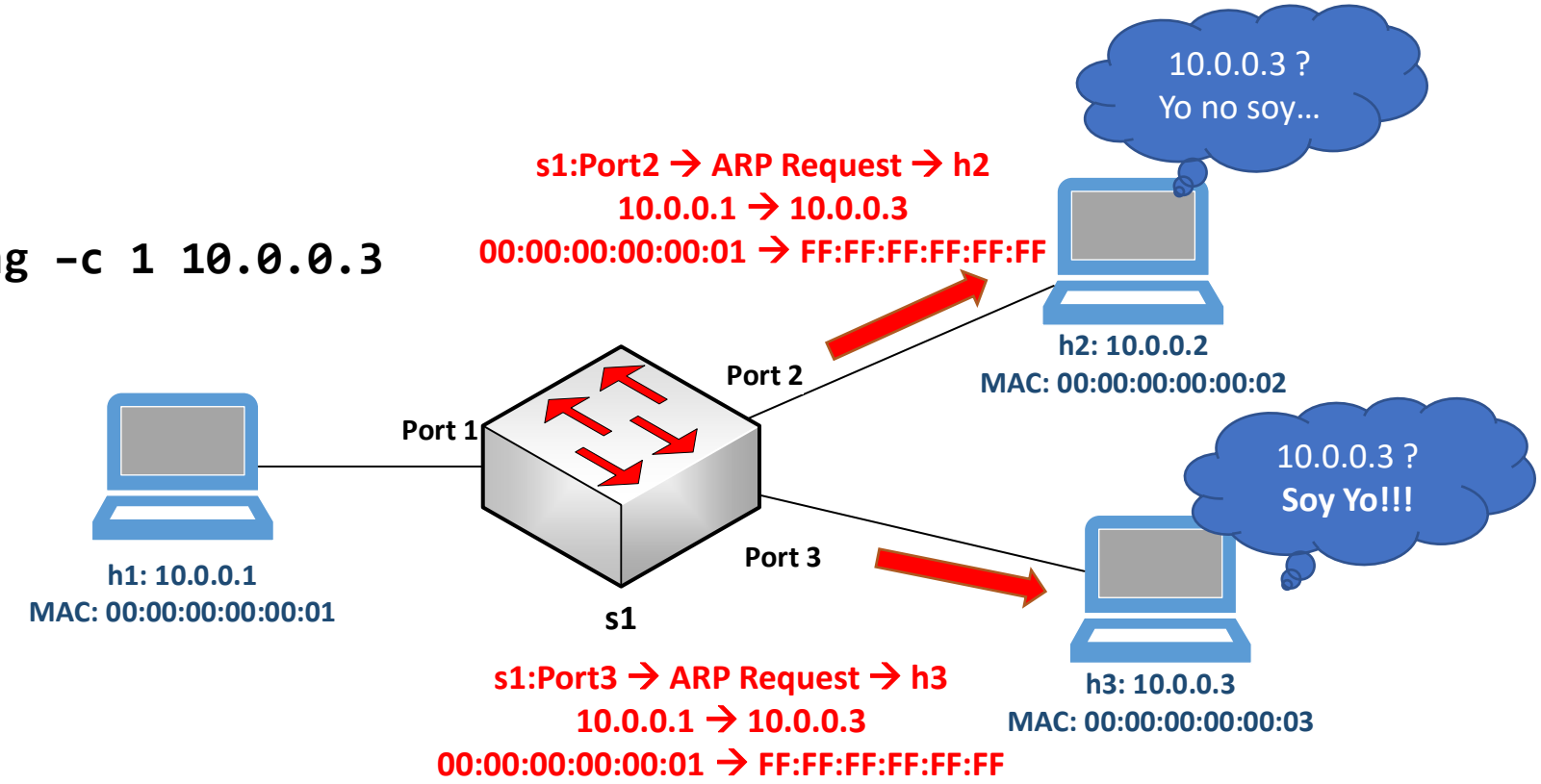
h1 ARP Table	
IP	MAC

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC

Responder ARP?

```
h1:~$ping -c 1 10.0.0.3
```



Switch Table	
MAC	Port
00:00:00:00:00:01	1

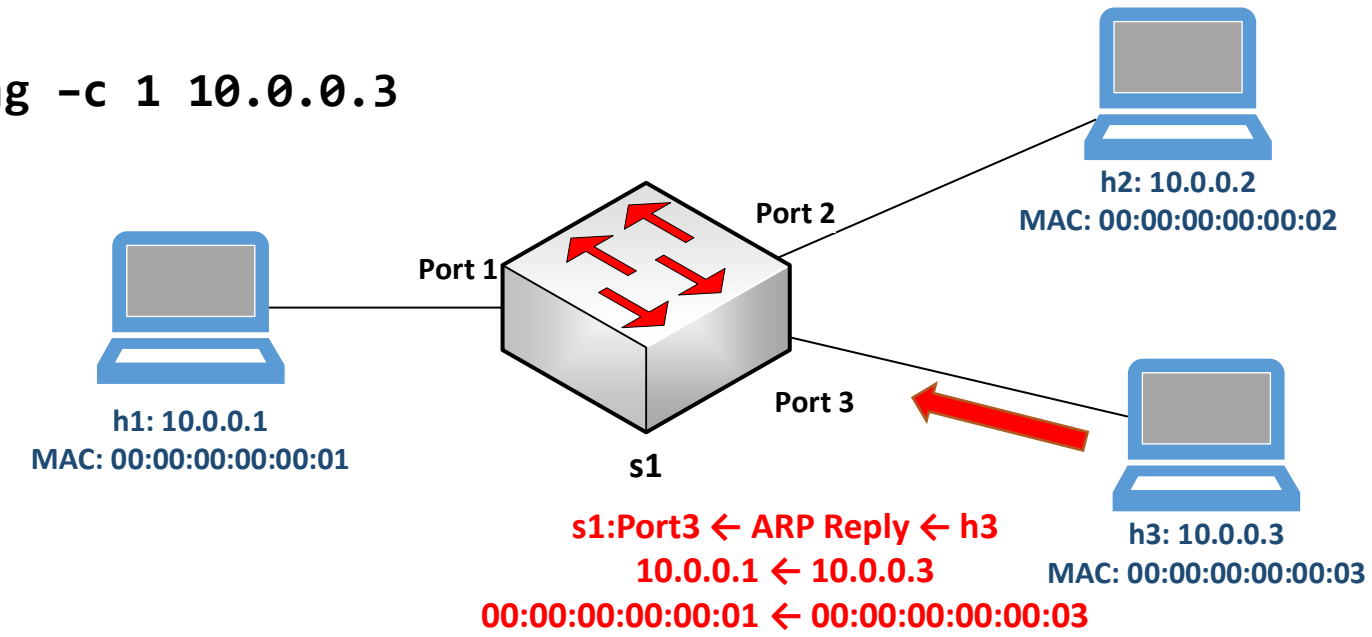
h1 ARP Table	
IP	MAC

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
10.0.0.1	00:00:00:00:00:01

ARP Reply

```
h1:~$ping -c 1 10.0.0.3
```



Switch Table	
MAC	Port
00:00:00:00:00:01	1

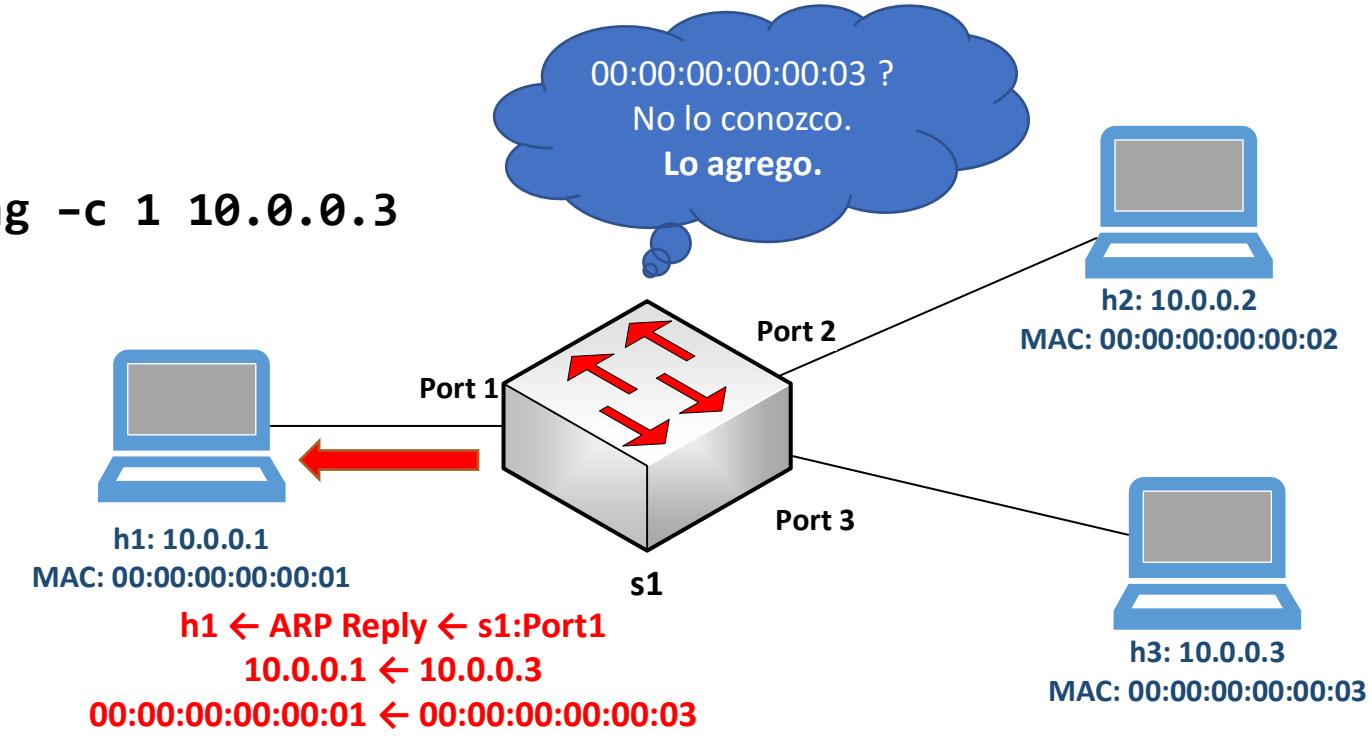
h1 ARP Table	
IP	MAC

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
10.0.0.1	00:00:00:00:00:01

Aprendizaje de MAC

```
h1:~$ping -c 1 10.0.0.3
```



Switch Table	
MAC	Port
00:00:00:00:00:01	1
00:00:00:00:00:03	3

h1 ARP Table	
IP	MAC

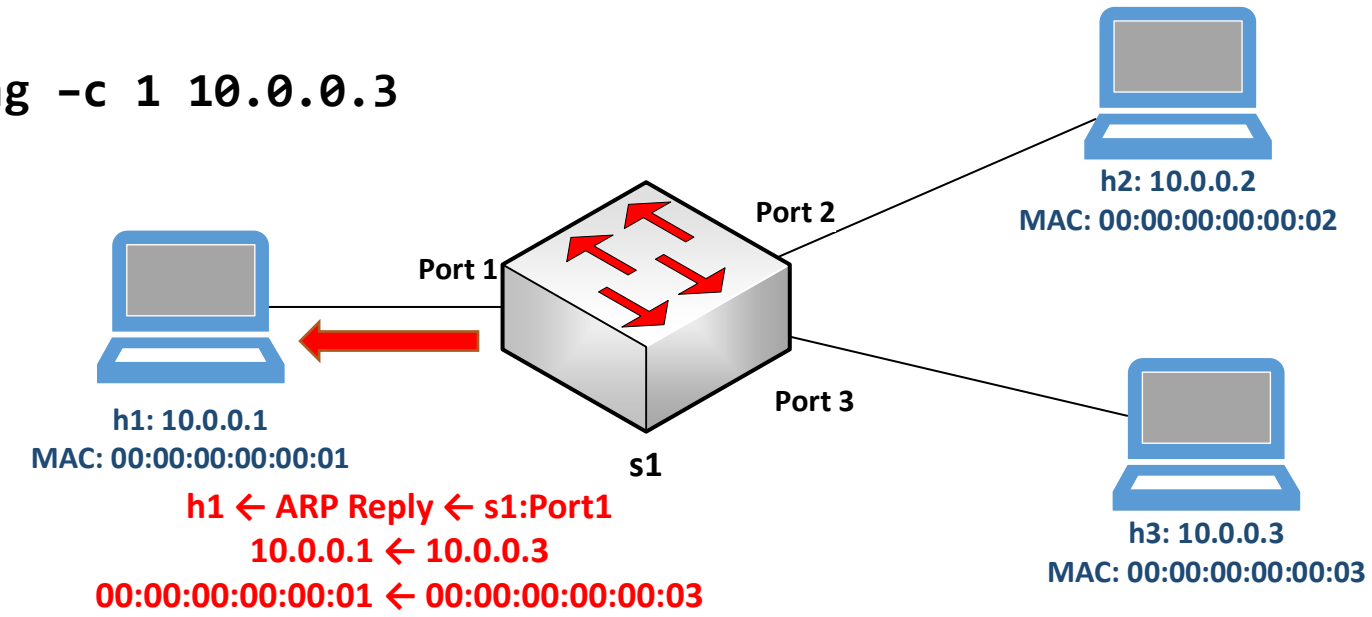
h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
10.0.0.1	00:00:00:00:00:01

Recepción de ARP Reply

Switch Table	
MAC	Port
00:00:00:00:00:01	1
00:00:00:00:00:03	3

```
h1:~$ping -c 1 10.0.0.3
```



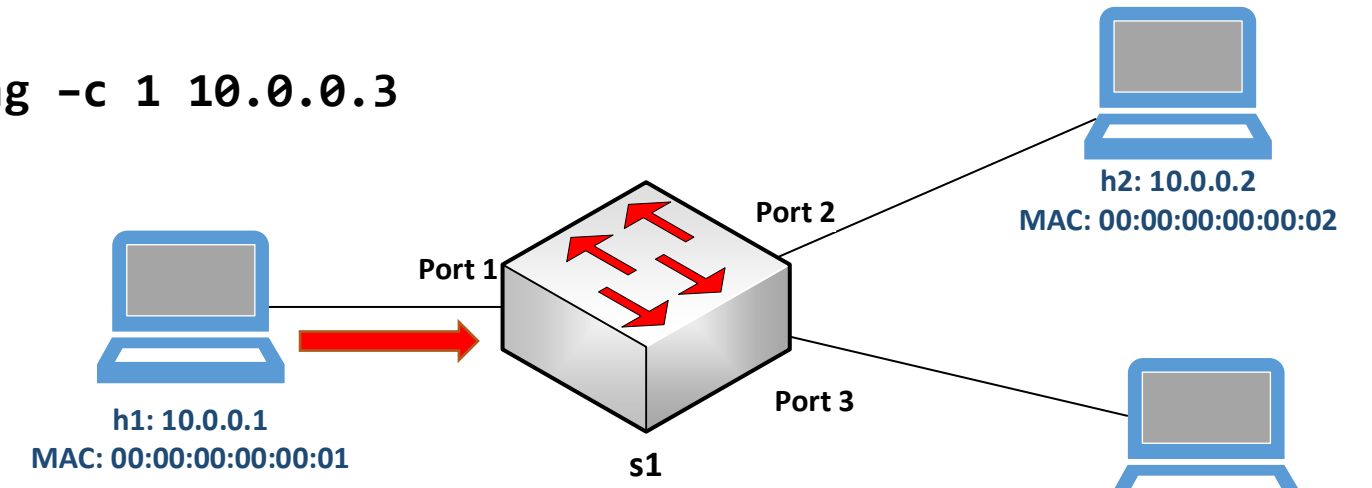
h1 ARP Table	
IP	MAC
10.0.0.3	00:00:00:00:00:03

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
10.0.0.1	00:00:00:00:00:01

ICMP Request

```
h1:~$ping -c 1 10.0.0.3
```



h1 → ICMP Echo Request → s1:Port1
10.0.0.1 → 10.0.0.3
00:00:00:00:00:01 → 00:00:00:00:00:03

Switch Table	
MAC	Port
00:00:00:00:00:01	1
00:00:00:00:00:03	3

h1 ARP Table	
IP	MAC
10.0.0.3	00:00:00:00:00:03

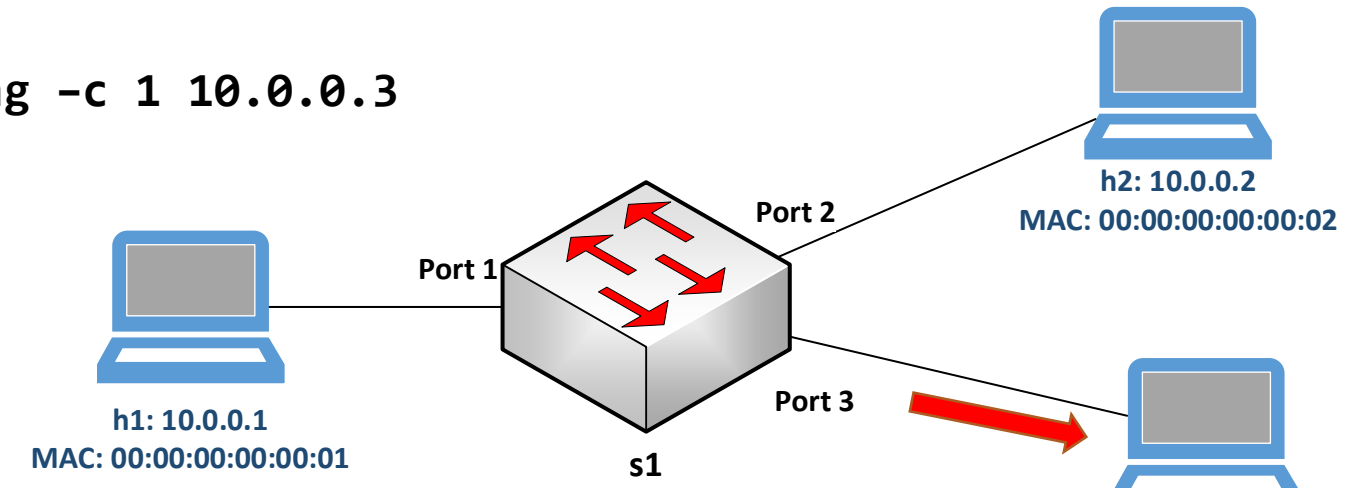
h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
10.0.0.1	00:00:00:00:00:01

Frame Forwarding

Switch Table	
MAC	Port
00:00:00:00:00:01	1
00:00:00:00:00:03	3

```
h1:~$ping -c 1 10.0.0.3
```



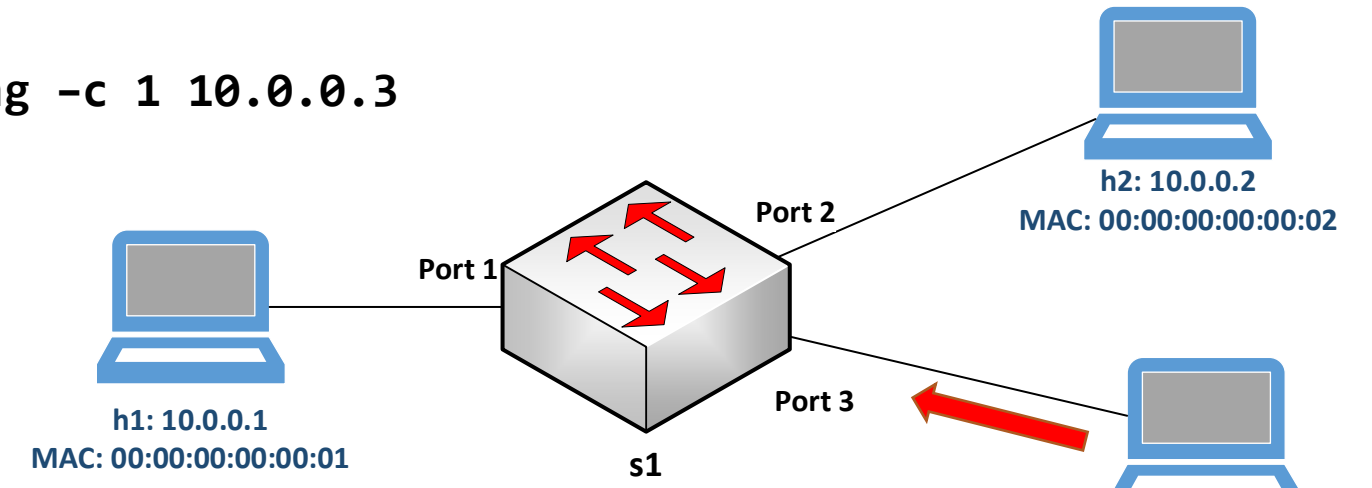
h1 ARP Table	
IP	MAC
10.0.0.3	00:00:00:00:00:03

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
10.0.0.1	00:00:00:00:00:01

ICMP Reply

```
h1:~$ping -c 1 10.0.0.3
```



s1:Port3 ← ICMP Echo Reply ← h3
10.0.0.1 ← 10.0.0.3
00:00:00:00:00:01 ← 00:00:00:00:00:03

Switch Table	
MAC	Port
00:00:00:00:00:01	1
00:00:00:00:00:03	3

h1 ARP Table	
IP	MAC
10.0.0.3	00:00:00:00:00:03

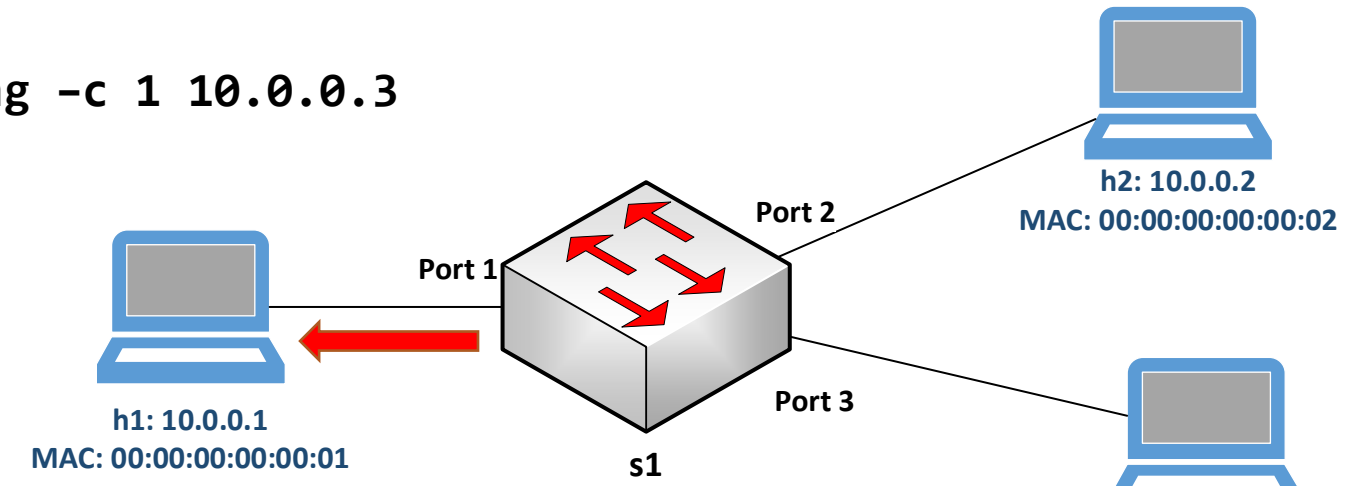
h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
10.0.0.1	00:00:00:00:00:01

Frame Forwarding

Switch Table	
MAC	Port
00:00:00:00:00:01	1
00:00:00:00:00:03	3

```
h1:~$ping -c 1 10.0.0.3
```



h1 ← ICMP Echo Reply ← s1:Port1
10.0.0.1 ← 10.0.0.3
00:00:00:00:00:01 ← 00:00:00:00:00:03

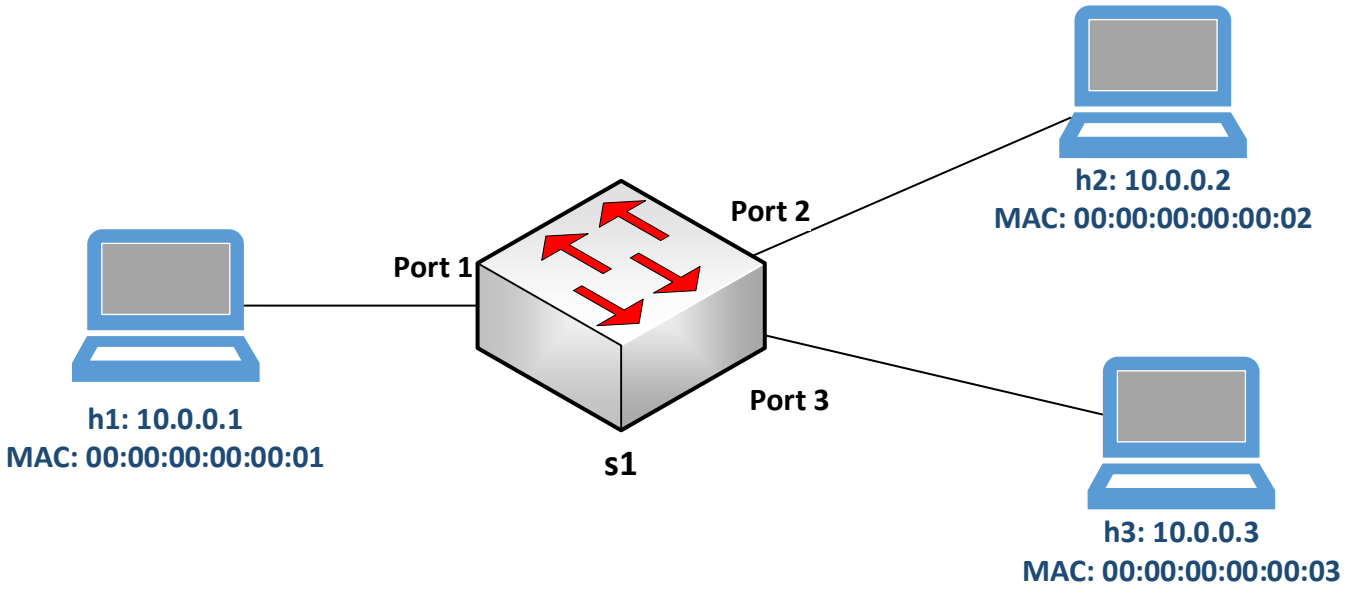
h1 ARP Table	
IP	MAC
10.0.0.3	00:00:00:00:00:03

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
10.0.0.1	00:00:00:00:00:01

Recepción ICMP Reply

```
h1:~$ping -c 1 10.0.0.3
64 bytes from 10.0.0.3: icmp_seq=1 ttl=128 time=15 min
```



ttl=128 No se decrementó (No pasó por ningún Router)

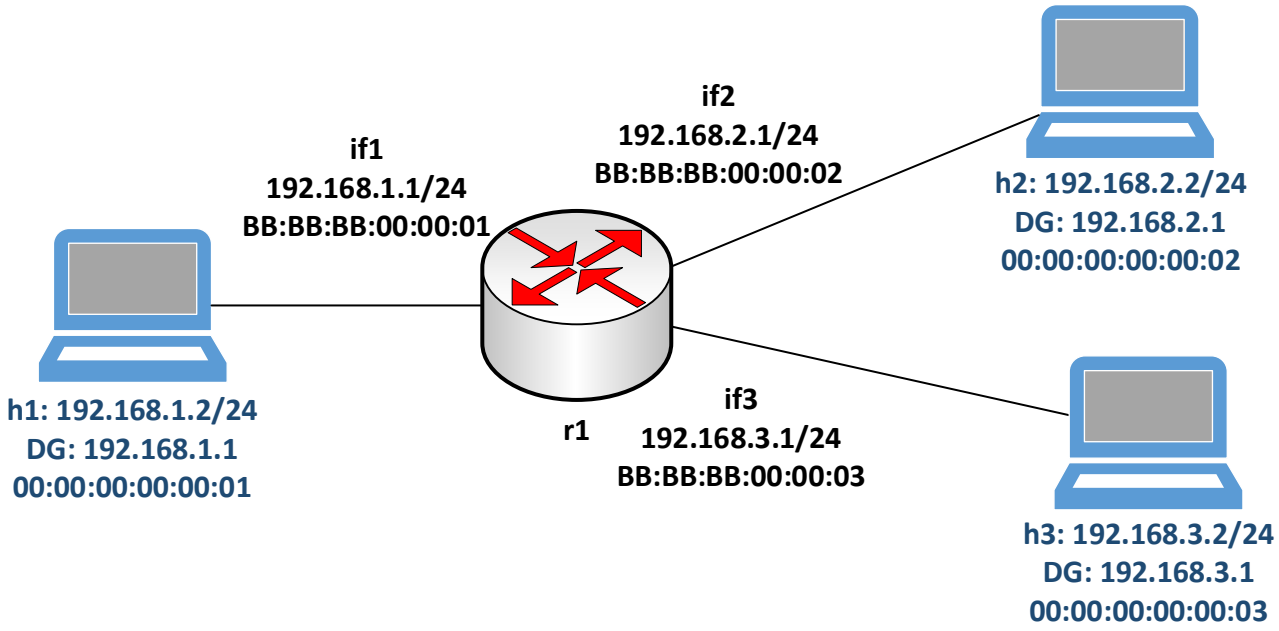
Switch Table	
MAC	Port
00:00:00:00:00:01	1
00:00:00:00:00:03	3

h1 ARP Table	
IP	MAC
10.0.0.3	00:00:00:00:00:03

h2 ARP Table	
IP	MAC

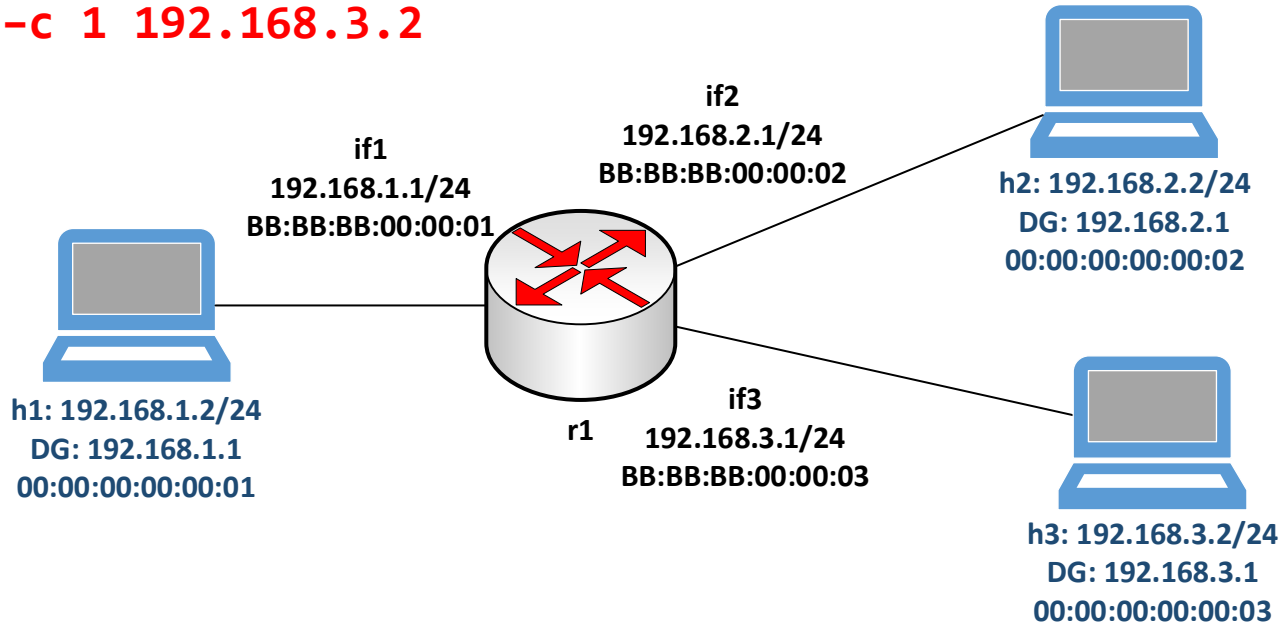
h3 ARP Table	
IP	MAC
10.0.0.1	00:00:00:00:00:01

Otra Topología: Router



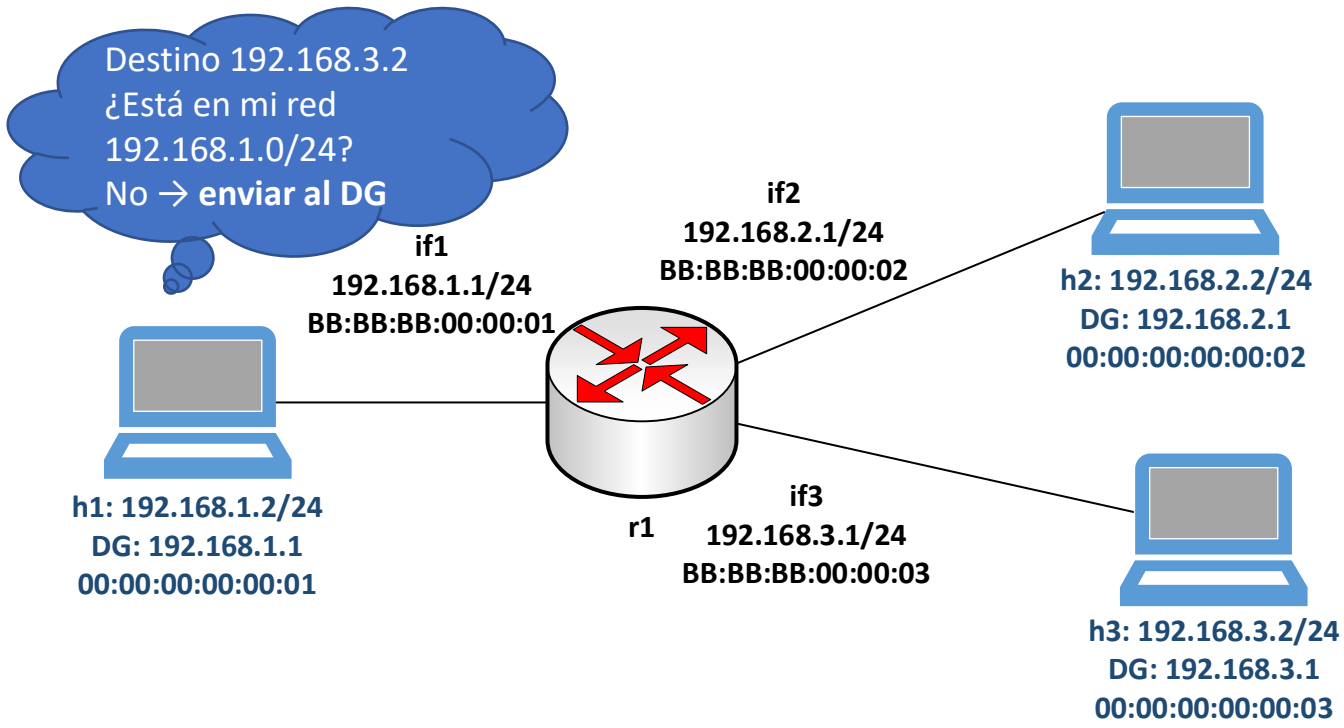
Ping

```
h1:~$ping -c 1 192.168.3.2
```



Tablas (casi vacías)

```
h1:~$ping -c 1 192.168.3.2
```



r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

Router r1 ARP Table		
IP	if	MAC

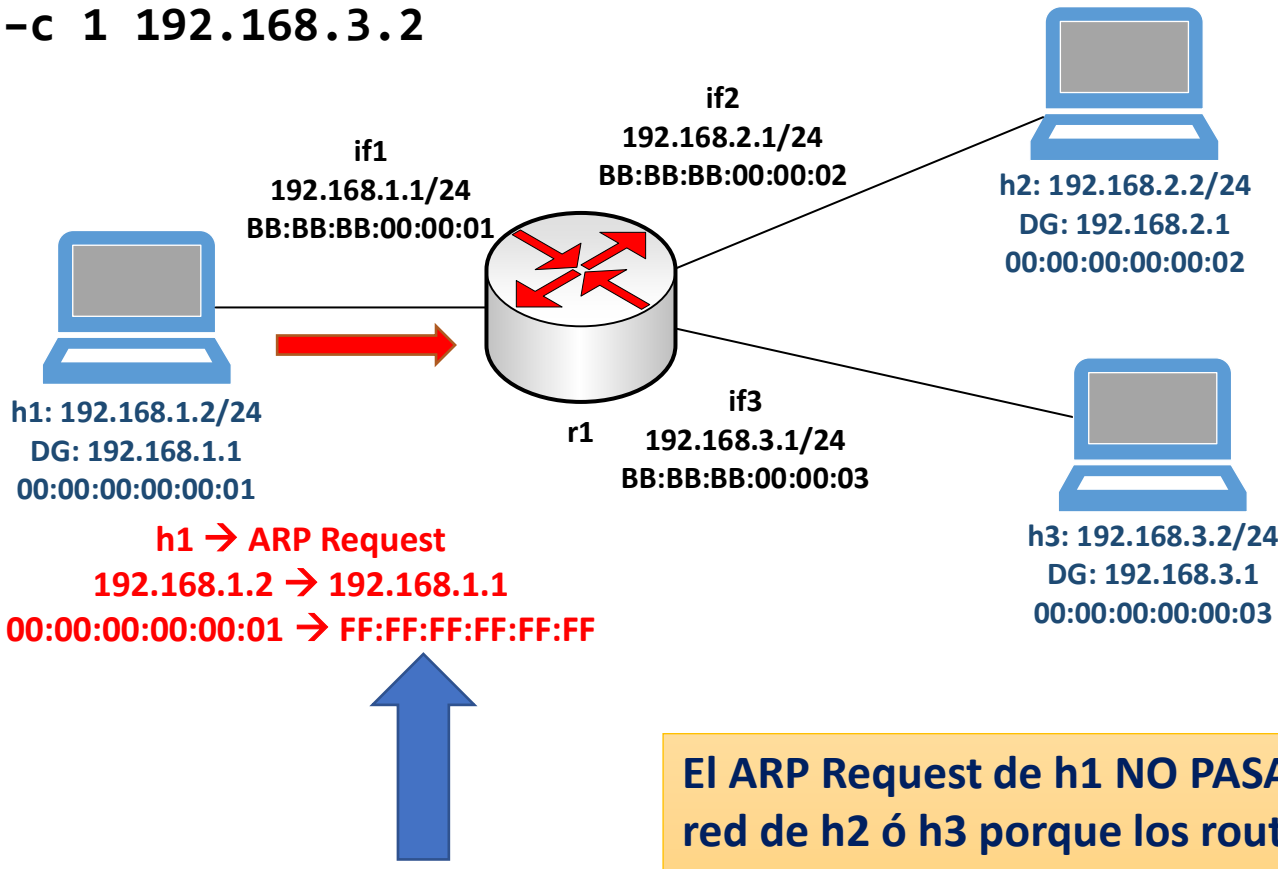
h1 ARP Table	
IP	MAC

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC

ARP Request

```
h1:~$ping -c 1 192.168.3.2
```



IP del DG
MAC Broadcast

El ARP Request de h1 NO PASA a la red de h2 ó h3 porque los routers no reenvían broadcasts.

r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

Router r1 ARP Table		
IP	if	MAC

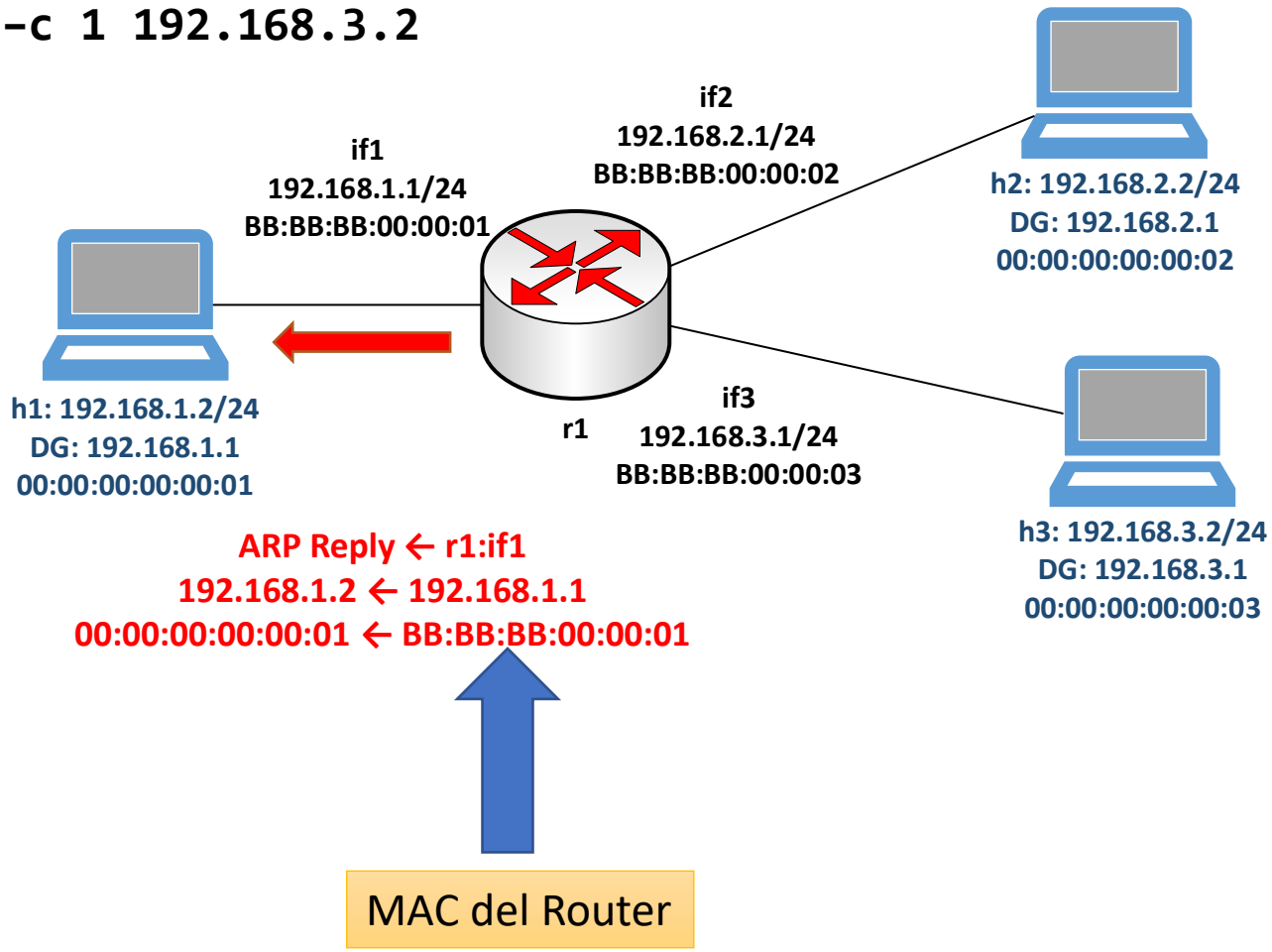
h1 ARP Table	
IP	MAC

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC

ARP Reply

```
h1:~$ping -c 1 192.168.3.2
```



r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

Router r1 ARP Table		
IP	if	MAC
192.168.1.2	1	00:00:00:00:00:01

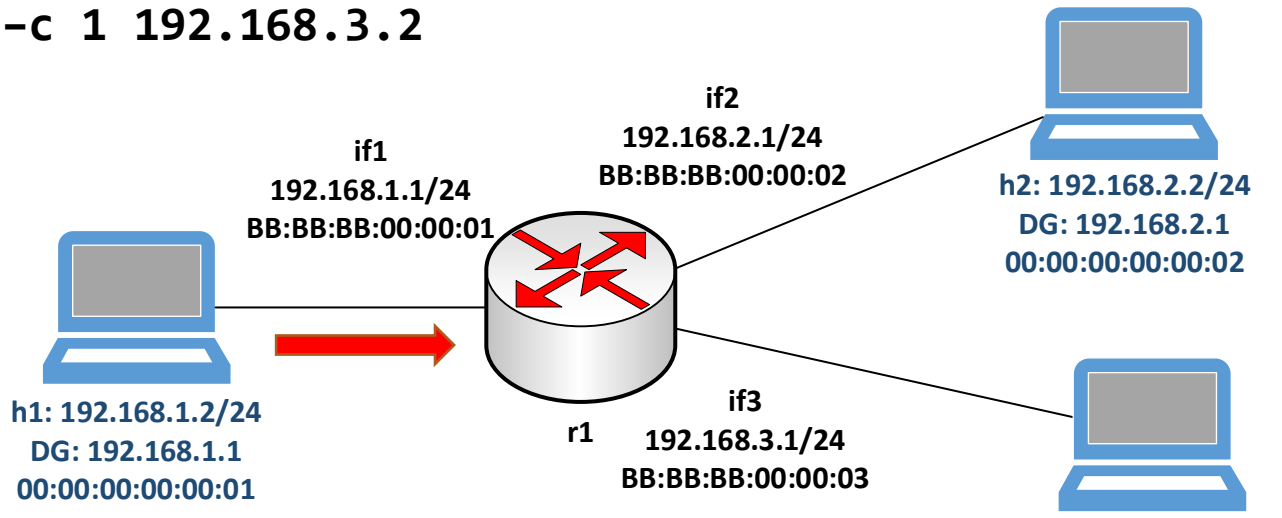
h1 ARP Table	
IP	MAC

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC

ICMP Echo Request

```
h1:~$ping -c 1 192.168.3.2
```



h1 → ICMP Echo Request
192.168.1.2 → 192.168.3.2
00:00:00:00:00:01 → BB:BB:BB:00:00:01

Destino:
IP de Host
MAC del Router

r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

Router r1 ARP Table		
IP	if	MAC
192.168.1.2	1	00:00:00:00:00:01

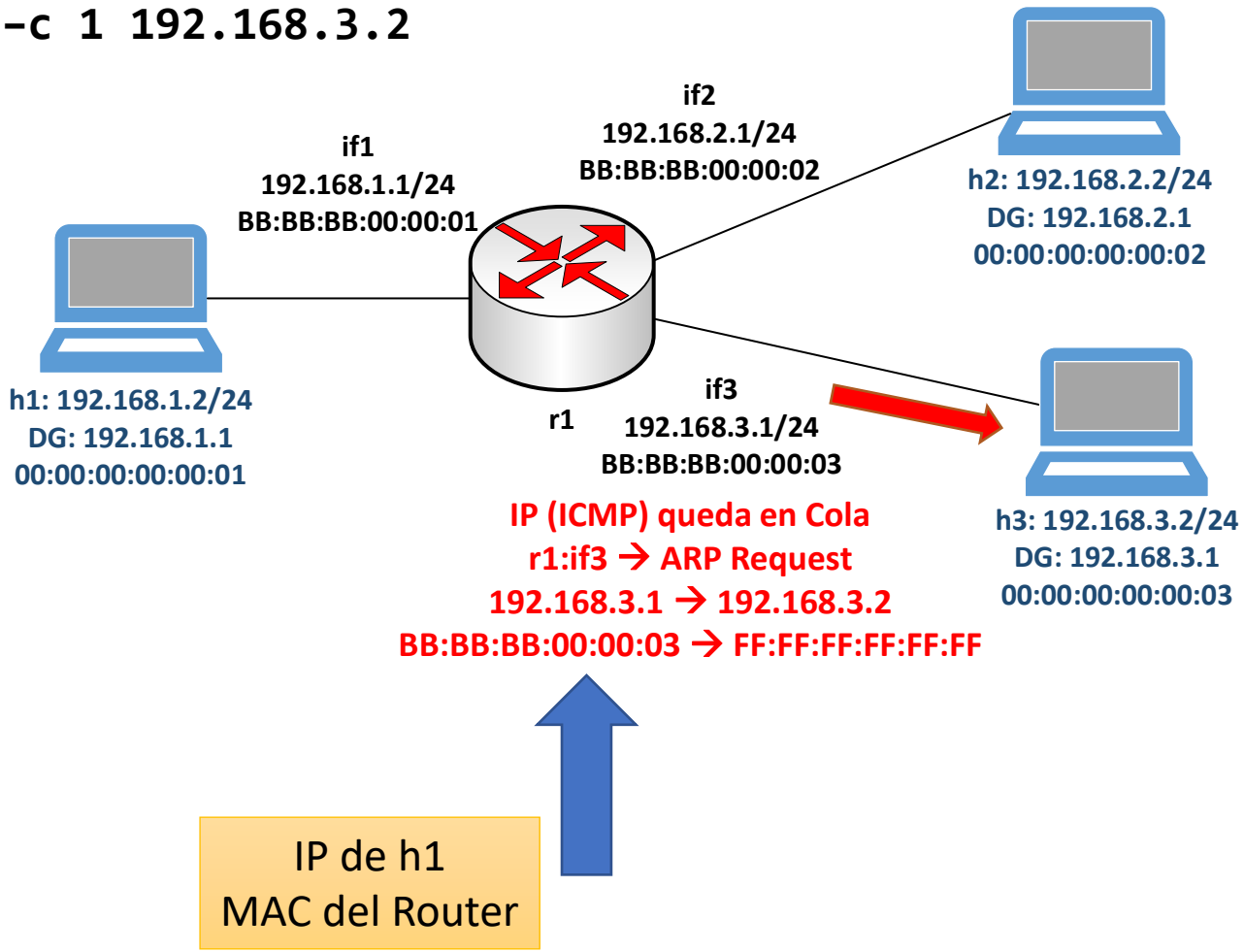
h1 ARP Table	
IP	MAC
192.168.1.1	BB:BB:BB:00:00:01

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC

Otro ARP Request

```
h1:~$ping -c 1 192.168.3.2
```



r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

Router r1 ARP Table		
IP	if	MAC
192.168.1.2	1	00:00:00:00:00:01

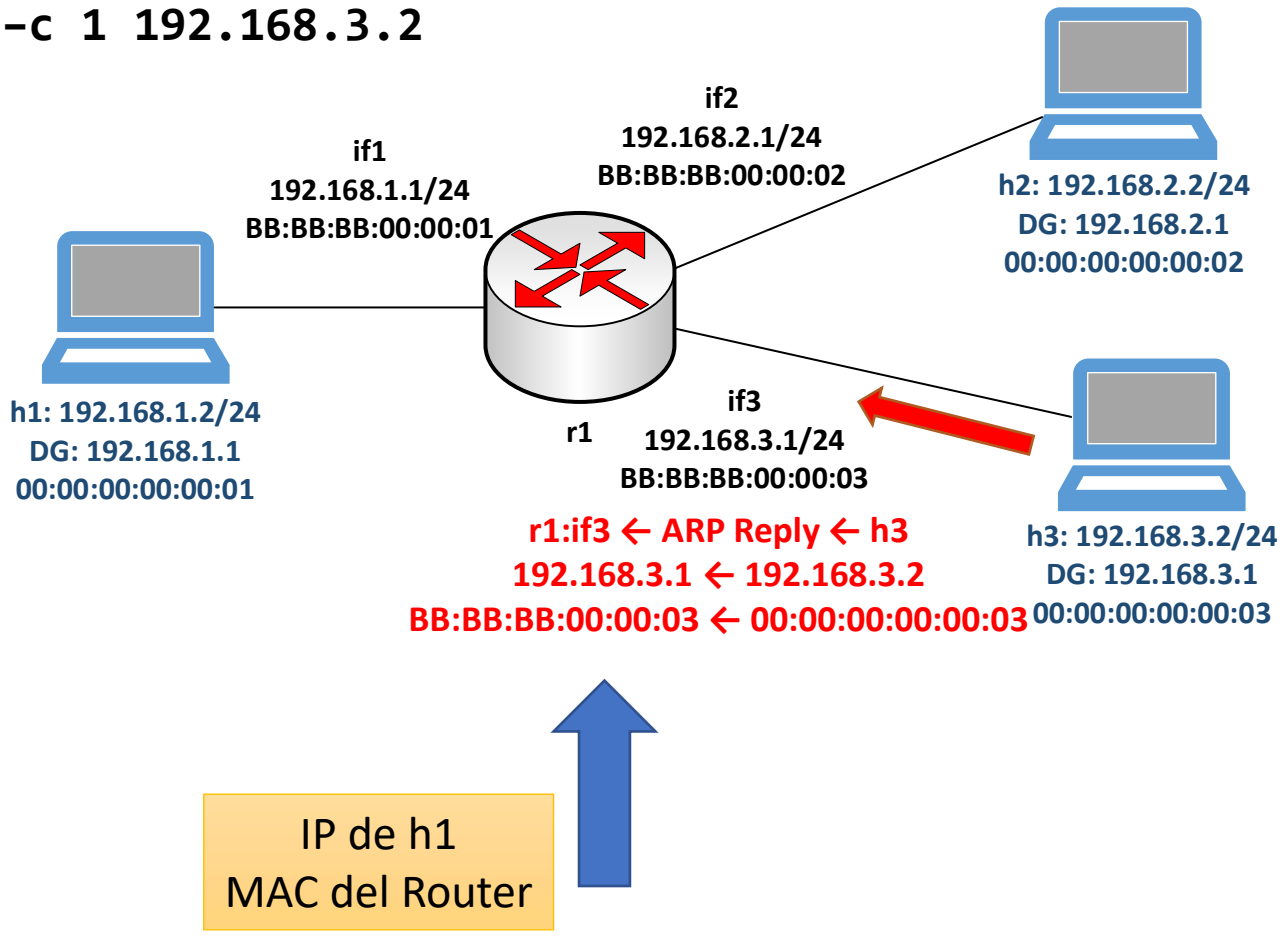
h1 ARP Table	
IP	MAC
192.168.1.1	BB:BB:BB:00:00:01

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC

Otro ARP Reply

```
h1:~$ping -c 1 192.168.3.2
```



r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

Router r1 ARP Table		
IP	if	MAC
192.168.1.2	1	00:00:00:00:00:01

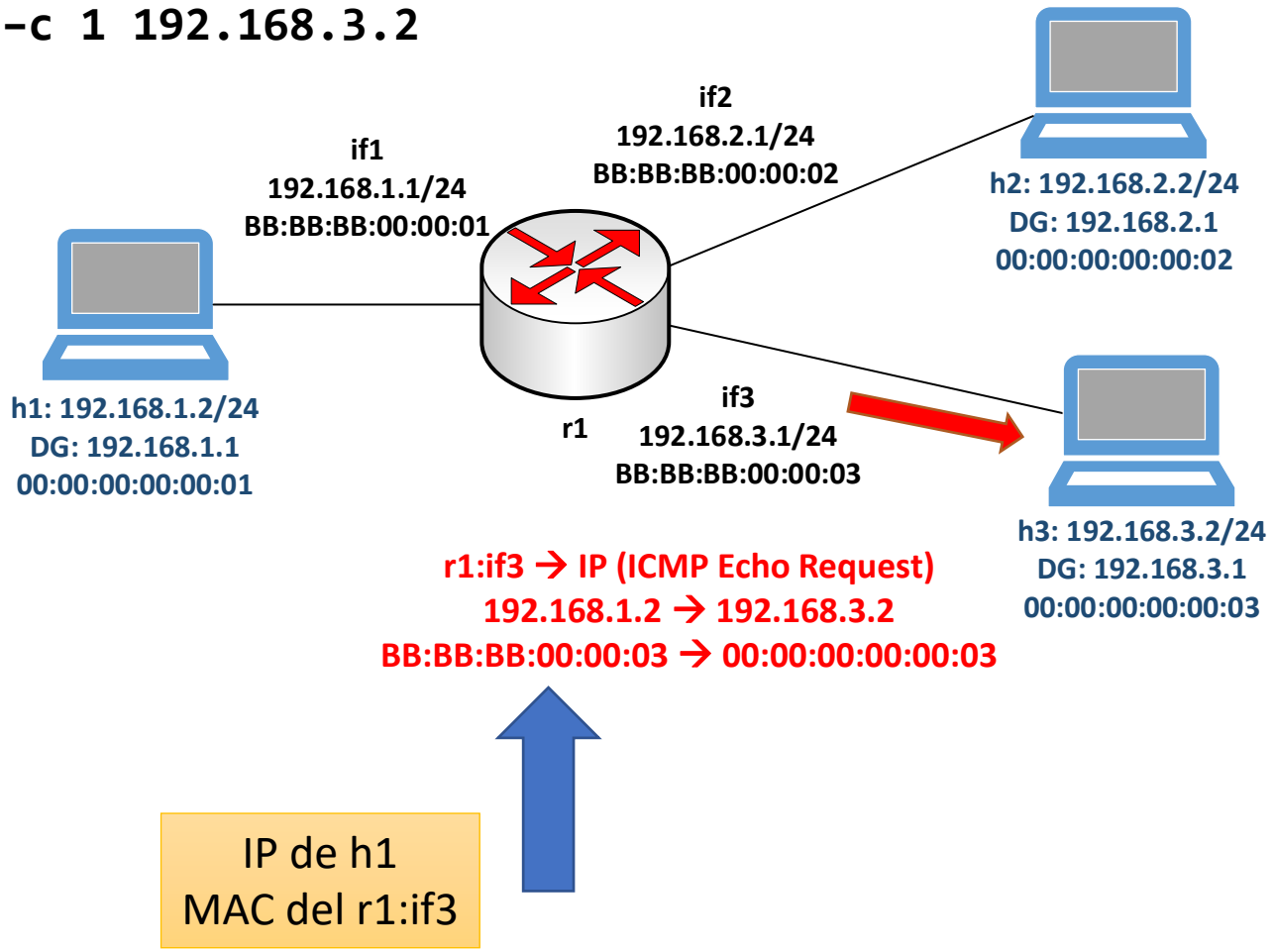
h1 ARP Table	
IP	MAC
192.168.1.1	BB:BB:BB:00:00:01

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
192.168.3.1	BB:BB:BB:00:00:03

IP Packet Forwarding - Capa 3 (IP)

```
h1:~$ping -c 1 192.168.3.2
```



r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

Router r1 ARP Table		
IP	if	MAC
192.168.1.2	1	00:00:00:00:00:01
192.168.3.2	3	00:00:00:00:00:03

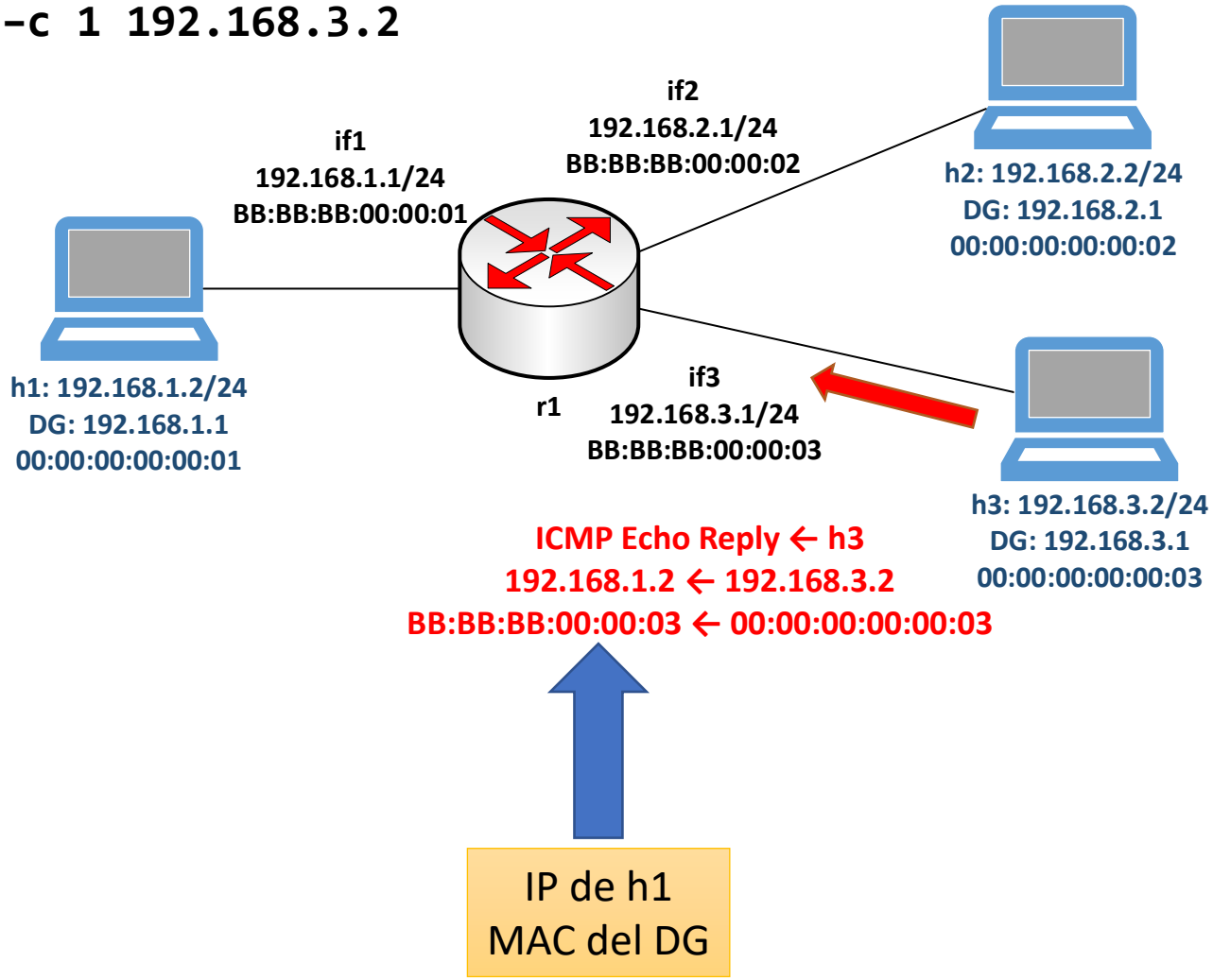
h1 ARP Table	
IP	MAC
192.168.1.1	BB:BB:BB:00:00:01

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
192.168.3.1	BB:BB:BB:00:00:03

ICMP Echo Reply

```
h1:~$ping -c 1 192.168.3.2
```



r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

Router r1 ARP Table		
IP	if	MAC
192.168.1.2	1	00:00:00:00:00:01
192.168.3.2	3	00:00:00:00:00:03

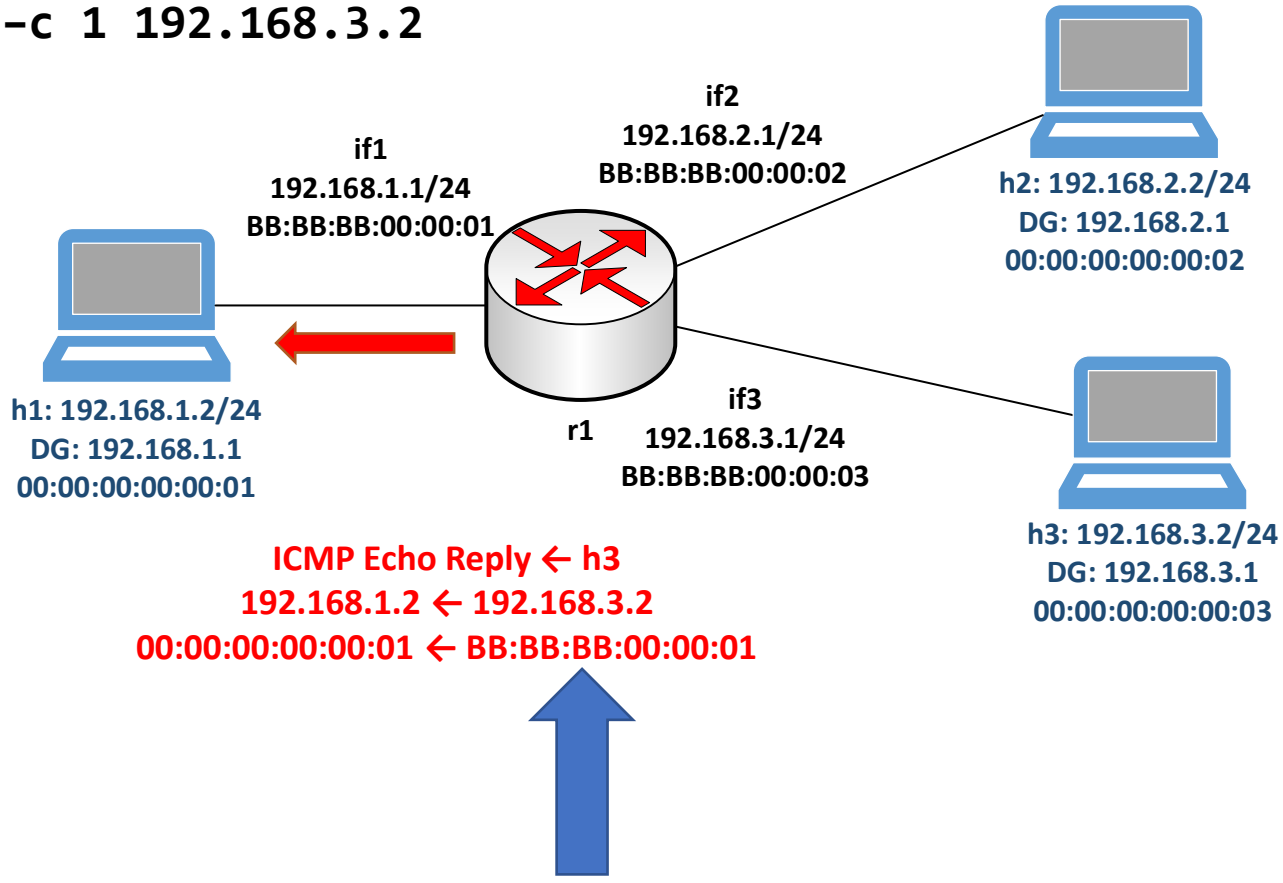
h1 ARP Table	
IP	MAC
192.168.1.1	BB:BB:BB:00:00:01

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
192.168.3.1	BB:BB:BB:00:00:03

IP Packet Forwarding

```
h1:~$ping -c 1 192.168.3.2
```



r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

Router r1 ARP Table		
IP	if	MAC
192.168.1.2	1	00:00:00:00:00:01
192.168.3.2	3	00:00:00:00:00:03

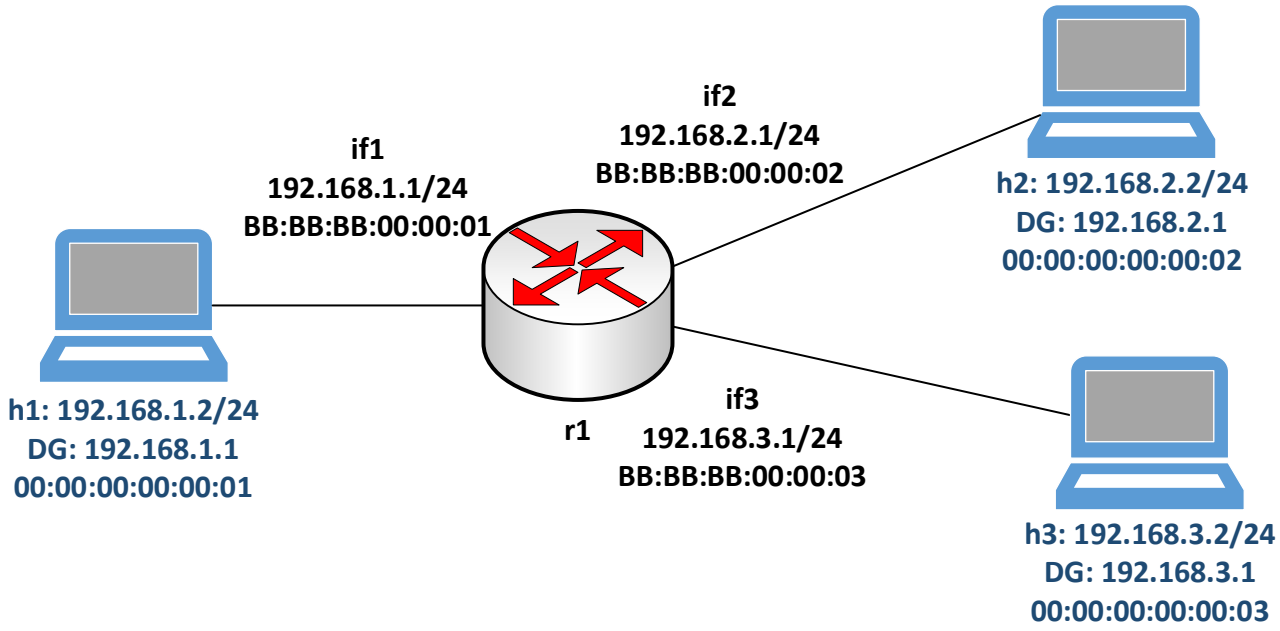
h1 ARP Table	
IP	MAC
192.168.1.1	BB:BB:BB:00:00:01

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
192.168.3.1	BB:BB:BB:00:00:03

Respondió el ping

```
h1:~$ping -c 1 192.168.3.2
64 bytes from 192.168.3.2: icmp_seq=1 ttl=127 time=10 min
```



ttl=127 Se decrementó! (al pasar por r1)

r1 Routing Table	
Subnet	Next-Hop
192.168.1.0/24	direct (if1)
192.168.2.0/24	direct (if2)
192.168.3.0/24	direct (if3)

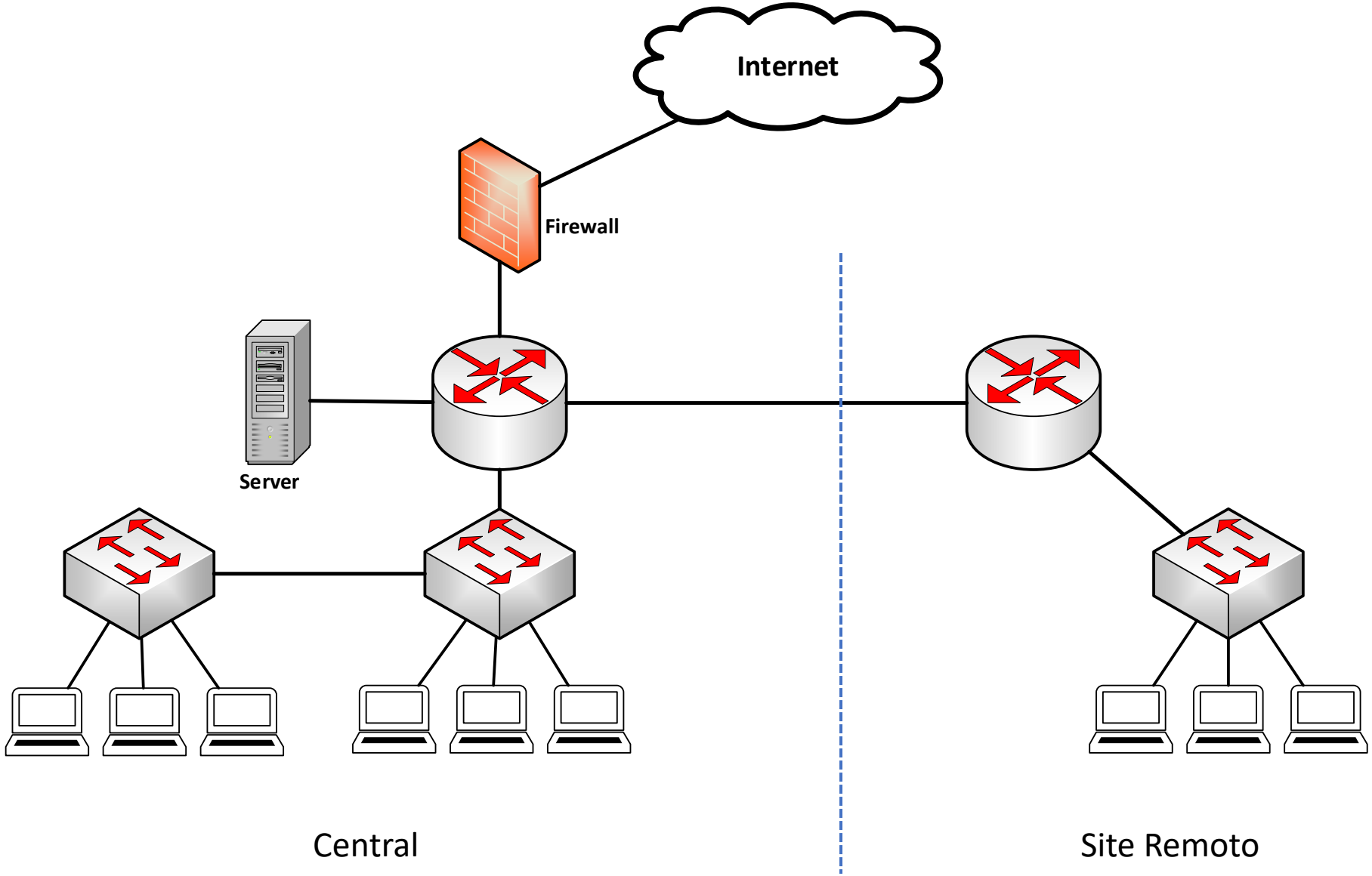
Router r1 ARP Table		
IP	if	MAC
192.168.1.2	1	00:00:00:00:00:01
192.168.3.2	3	00:00:00:00:00:03

h1 ARP Table	
IP	MAC
192.168.1.1	BB:BB:BB:00:00:01

h2 ARP Table	
IP	MAC

h3 ARP Table	
IP	MAC
192.168.3.1	BB:BB:BB:00:00:03

Topología Chica



Topología Jerárquica

Internet

Firewalls

Core Layer
Switchs de Muy Alta Velocidad

Distribution Layer
Switchs de Capa 3 (Routers)

Access Layer
Switchs – VLANS – Muchos ports

Usuarios

